

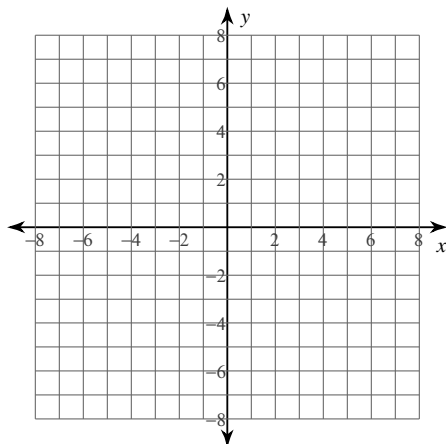
For each problem, find the volume of the solid that results when the region enclosed by the curves is revolved about the given axis. You may use the provided graph to sketch the curves and shade the enclosed region.

5) $y = x^2 - 2$

$y = -2$

$x = 2$

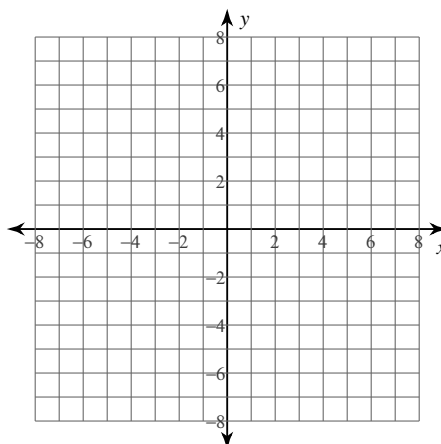
Axis: $y = -2$



6) $x = \sqrt{y} + 3$

$x = \frac{y}{2} + 3$

Axis: $x = 1$



Critical thinking questions:

7) Use the method of disks to derive the formula for the volume of a sphere of radius r .

8) A 6 cm diameter drill bit is used to drill a cylindrical hole through the middle of a sphere of radius 5 cm. What is the volume of the resulting object?